

Kamal Swarnkar

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SKILLS

Languages:	Python, C
Tools:	Git, Linux, Docker, SolidWorks, OpenRocket, VSCode
Libraries & ML:	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, XGBoost
Frameworks:	Django, Express.js, React
Databases:	PostgreSQL, MongoDB
Soft Skills:	Self-motivated, Articulate, Organized, Analytical, Leadership

PROJECTS

AI Career Lab | GitHub | Live Demo

Mar' 2026

- Architected an ML-based career guidance platform using TF-IDF and multi-class classification, achieving **86.58% accuracy** with **1.23% overfitting gap**.
- Evaluated RF, SVM, and LR using precision, recall, and F1-score; best model achieved **86.63% precision**, **86.58% recall**, and **86.56% F1-score** on unseen data.
- Engineered a robust training pipeline with explicit train-test gap analysis (RF: **4.66%**, SVM: **1.66%**, LR: **1.23%**) to ensure strong generalization and avoid overfitting.
- Developed a modular **Django app** with real-time inference for career prediction and **skill gap analysis**.
- **Tech Stack:** Python, Django, Scikit-learn, Pandas, NumPy.

Resumix - Resume Analysis & Role Prediction Platform | GitHub | Live Demo

Feb' 2026

- Built a production-oriented **Django + DRF platform** that authenticates users with **JWT**, ingests PDF/DOCX resumes, parses text, and delivers end-to-end resume analysis through a unified web UI and REST APIs
- Engineered a weighted scoring pipeline (TF-IDF, skills, experience, ATS) with **automated gap detection** and recommendations.
- Benchmarked **6 ML models** (LR, SVM, RF, XGBoost, KNN, DT) using PR-F1, ROC-AUC, MCC; deployed best model.
- Implemented **RBAC and admin analytics APIs** to track users, resumes, and system usage.
- **Tech Stack:** Python, Django, PostgreSQL, JWT, Scikit-learn, XGBoost, spaCy, NLTK, Celery, Redis.

FinTrackAI | GitHub | Live Demo

Jul' 2025

- Spearheaded the development of an AI-driven expense management system leveraging Generative AI for automated transaction categorization, achieving **92% classification accuracy** while reducing manual logging effort by **65%**.
- Devised a data processing pipeline to **handle unstructured financial inputs**, improving categorization consistency and reducing user-level errors by **48%** through **intelligent normalization** and preprocessing.
- Crafted an interactive **analytics dashboard** delivering insights on budgets, spending patterns, and savings trends, enhancing user financial awareness and engagement.
- Established a **scalable full-stack architecture** with modular API design and real-time data updates for seamless financial tracking and analysis.
- **Tech Stack:** MongoDB, Express.js, Node.js, React.js.

TRAINING

FLAMES '25 MERN with Gen AI | W3grads | Certificate

Jul' 2025

- Addressed major development bottlenecks—skill gaps, AI inconsistencies, database lag, and deployment issues; streamlined workflows and optimized processes, leading to faster development, reliable AI performance, and stable deployments.
- Encountered collaboration hurdles like merge conflicts and time-zone delays; established a strict Git workflow and daily SCRUMs, which reduced conflicts and kept the team aligned.

Youth Research Program | Resolute Lab | Certificate

Mar' 2025

- Applied aerospace engineering concepts to design and simulate rocket systems using OpenRocket and SolidWorks; performed propulsion analysis, payload integration studies, and structural feasibility simulations.
- Collaborated in a multidisciplinary team to design a high-power rocket with simulated payload, optimizing parameters via iterative simulations and maintaining precise technical documentation.

CERTIFICATES

Master Generative AI & Generative AI tools (Infosys) Certificate	Aug' 2025
The Bits and Bytes of Computer Networking (Google) Certificate	Sep' 2024
Computer Communications (University of Colorado) Certificate	Sep' 2024

Achievements

- Solved 50+ DSA Problems on LeetCode.
- Participated 20+ Technical and Industry workshops.

EDUCATION

Lovely Professional University	Phagwara, Punjab
Bachelor of Technology - Computer Science and Engineering – CGPA: 8.87	Aug' 2023 – Present
Central Academy Senior Secondary School	Bhilwara, Rajasthan
Intermediate – 91%	Apr' 2021 – Mar'2022
Matriculation – 89%	Apr' 2019 – Mar'2020